



## **Lab Task 06**

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|---------------------|---------------------------|
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| <b>Subject</b>      | <b>SCD</b>                |

```
*CarRentalSystem.java ×
1 package project1;
2
3 /*
4      Legal Notice
5      Copyright (c) 2025 Mardan Motors. All rights reserved.
6      This software is the proprietary information of Mardan Motors.
7      Unauthorized use, copying, or distribution is strictly prohibited.
8  */
9
10 public class CarRentalSystem {
11
12     // Fields for car details
13     private String carModel;
14     private String carLicensePlate;
15     private int carYear;
16
17     // Fields for rental duration
18     private int rentalDurationDays;
19
20     // Fields for customer information
21     private String customerName;
22     private String customerPhoneNumber;
23
24
25
26     public CarRentalSystem(String carModel, String carLicensePlate, int carYear, int rentalDurationDays,
27                             String customerName, String customerPhoneNumber) {
28
29         this.carModel = carModel;
30         this.carLicensePlate = carLicensePlate;
31         this.carYear = carYear;
32         this.rentalDurationDays = rentalDurationDays;
33         this.customerName = customerName;
34         this.customerPhoneNumber = customerPhoneNumber;
35     }
36
37
38
39     // Getters and Setters for car details
40     public String getCarModel() {
41         return carModel;
42     }
43
44     // TODO: Add validation for car model (non-null, non-empty)
45     public void setCarModel(String carModel) {
46         this.carModel = carModel;
47     }
48
49     public String getCarLicensePlate() {
50         return carLicensePlate;
51     }
52
53     // TODO: Add license plate format validation
54     public void setCarLicensePlate(String carLicensePlate) {
55         this.carLicensePlate = carLicensePlate;
56     }
57
58     public int getCarYear() {
59         return carYear;
60     }
61 }
```

```

61
62 // TODO: Validate year range (e.g., 1900-current year+1)
63 public void setCarYear(int carYear) {
64     this.carYear = carYear;
65 }
66
67
68 public int getRentalDurationDays() {
69     return rentalDurationDays;
70 }
71
72 // TODO: Add validation for positive rental duration
73 public void setRentalDurationDays(int rentalDurationDays) {
74     this.rentalDurationDays = rentalDurationDays;
75 }
76
77 // Getters and Setters for customer information
78 public String getCustomerName() {
79     return customerName;
80 }
81
82 // TODO: Add name validation (non-null, non-empty)
83 public void setCustomerName(String customerName) {
84     this.customerName = customerName;
85 }
86
87 public String getCustomerPhoneNumber() {
88     return customerPhoneNumber;
89 }
90
91 // TODO: Implement phone number format validation
92 public void setCustomerPhoneNumber(String customerPhoneNumber) {
93     this.customerPhoneNumber = customerPhoneNumber;
94 }
95
96
97
98
99 public void printDetails() {
100     System.out.println("Car Details:");
101     System.out.println("Model: " + carModel);
102     System.out.println("License Plate: " + carLicensePlate);
103     System.out.println("Year: " + carYear);
104     System.out.println("\nRental Duration:");
105     System.out.println("Days: " + rentalDurationDays);
106     System.out.println("\nCustomer Information:");
107     System.out.println("Name: " + customerName);
108     System.out.println("Phone Number: " + customerPhoneNumber);
109 }
110
111
112
113 // TODO: Add method to calculate rental cost based on duration and car type
114 // TODO: Add method to save rental information to database/file
115 // TODO: Add method to load rental information from database/file
116
117 public static void main(String[] args) {
118     // Create an instance of the CarRentalSystem
119     CarRentalSystem rental = new CarRentalSystem(
120         "Toyota Camry",
121         "ABC-1234",
122         2020,
123         7,

```

```

123         /,
124         "John Doe",
125         "123-456-7890"
126     );
127
128
129
130
131     System.out.println("Initial Rental Details:");
132     rental.printDetails();
133
134
135     rental.setRentalDurationDays(10);
136     rental.setCustomerName("Jane Smith");
137
138
139     System.out.println("\nUpdated Rental Details:");
140     rental.printDetails();
141 }
142 }

```

## Output

```

Initial Rental Details:
Car Details:
Model: Toyota Camry
License Plate: ABC-1234
Year: 2020

Rental Duration:
Days: 7

Customer Information:
Name: John Doe
Phone Number: 123-456-7890

Updated Rental Details:
Car Details:
Model: Toyota Camry
License Plate: ABC-1234
Year: 2020

Rental Duration:
Days: 10

Customer Information:
Name: Jane Smith
Phone Number: 123-456-7890

```

```

1 package project1;
2
3 public class ZakatCalculator {
4
5     private static final double ZAKAT_PERCENTAGE=0.025;
6
7     public double calculate_zakat(double total_savings) {
8
9
10    // Amplification: It's crucial to validate user input before performing financial calculations.
11    // This prevents logical errors and ensures the integrity of the zakat calculation.
12    if (total_savings < 0) {
13        throw new IllegalArgumentException("Total saving cannot be negative.");
14    }
15
16        return total_savings * ZAKAT_PERCENTAGE;
17    }
18
19
20    public static void main(String[] args) {
21
22
23        ZakatCalculator calculator = new ZakatCalculator();           // Create an instance of the ZakatCalculator
24
25
26        double total_savings = 10000.0;                               // Example total savings amount
27        double zakat_amount = calculator.calculate_zakat(total_savings); // Calculate at based on the given savings
28
29
30        System.out.println("Total savings: $" + total_savings);
31        System.out.println("Zakat payable: $" + zakat_amount);
32    } }

```

## Output

```

Total savings: $10000.0
Zakat payable: $250.0

```